Course Project

The primary objective of the course project is to give you hands-on experiences in robot learning. For the project, you are expected to have a robot, at least in the simulator. If you can access to a real robot, using a real robot is encouraged. You are expected to form a team of 2 or 3 students to work on the project, and grades will be calibrated by team size.

Suggested Topics

Here is a list of suggested project topics. You are welcome to discuss with instructor and TA if you have other interesting topics you want to work on.

- Perception
  - Representation learning for planning and control; 3D/2D;
  - Incorporating large foundation model (e.g. DINO, SAM, Vision-Language Model)
- Dynamics
  - Model learning; Rigid / deformable / articulated / granular;
  - Comparison with physics-based simulators
- Planning & Control
  - Model-free RL; Model-based RL;
  - Model-predictive control (MPC)
  - Task and motion planning (TAMP)
  - LLM-guided planning
- System Integration & Application
  - Mobile / table-top manipulation
  - Dexterous manipulation; Bi-manual manipulation
  - Home robots
  - Agriculture robots
  - Medical robots
  - Warehouse robots

Grading Policy

The course project is worth 45% of the total grade. The following shows the breakdown:

- Project Proposal (5%). Due Feb 12 11:59PM
- Milestone (10%). Due March 26 11:59PM
- Project Report (15%). Due May 11 11:59PM
- Project Presentation (15%). Week 15&16

Project Proposal (5%)

Your project proposal report should be max 2 pages using the RSS template in LaTeX. The following is a suggested structure for your report:

- Title, Author(s)
- **Problem Overview (2%):** What is the problem that you will be investigating? Why is it interesting?
- **Literature Review (2%):** Describe important related work and their relevance to your project.
- **Proposed Method and Experiments (2%):** What method or algorithm are you proposing? You don't have to have an exact answer at this point, but you should have a general sense of how you will approach the problem you are working on. How will you evaluate your results, both quantitatively and qualitatively?
- **Milestone and Timeline (2%):** What are intermediate milestones you plan to accomplish? What is the final goal you plan to reach? What is the timeline for the project?
- **Team Background (2%):** What is your team's background? How is it related to your project?

Submission: Canvas

Milestone (10%)

Your project milestone report should be max 3 pages using the RSS template in LaTeX. The following is a suggested structure for your report:

- **Title, Author(s)**
- **Introduction (2%):** Introduce your problem and the overall plan for approaching your problem
- **Problem Statement (2%):** Describe your problem precisely specifying the dataset to be used, expected results and evaluation
- **Literature Review (2%):** Describe important related work and their relevance to your project
- **Technical Approach (2%):** Describe the methods you intend to apply to solve the given problem
- **Intermediate/Preliminary Results (2%):** State and evaluate your results up to the milestone

Submission: Canvas

Project Report (15%)

Your final write-up is required to be between 6-8 pages (8 pages max) using the RSS template in LaTeX. The following is a suggested structure for your report:

- **Title, Author(s)**
- **Abstract (1.5%):** Briefly describe your problem, approach, and key results. Should be no more than 300 words.
- **Introduction (1.5%):** Describe the problem you are working on, why it's important, and an overview of your results
- **Related Work (1.5%):** Discuss published work that relates to your project. How is your approach similar or different from others?
- **Methods (4.5%):** Discuss your approach for solving the problems that you set up in the introduction. Why is your approach the right thing to do? Did you consider alternative approaches? You should demonstrate that you have applied ideas and skills built up during the quarter to tackling your problem of choice. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.
- **Experiments (4.5%):** Discuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate your experimental results.

- **Conclusion (0.75%):** Summarize your key results - what have you learned? Suggest ideas for future extensions or new applications of your ideas.
- **Writing / Formatting (0.75%):** Is your paper clearly written and nicely formatted?
- **Supplementary Material** , not counted toward your 6-8 page limit and submitted as a separate file. Your supplementary material might include:
  - Source code (if your project proposed an algorithm, or code that is relevant and important for your project).
  - Cool videos, interactive visualizations, demos, etc.

  Examples of things to not put in your supplementary material:
  - The entire PyTorch/TensorFlow Github source code.
  - Any code that is larger than 10 MB.
  - Model checkpoints.
  - A computer virus.

Submission: Canvas

Project Presentation (15%)

You will have an opportunity to present your awesome work to the instructor and other students in the last three classes. Here are information about project presentation.

- **Time:** 10 minutes for presentation and 2 minutes for Q&A.
- **Contents:** similar to project report.
- **Grading scheme:**
  - Review prior works and describe why the problem is interesting (3%)
  - Describe methods correctly and clearly (3%)
  - Present experimental results and analyze results (3%)
  - Discuss future research directions (3%)
  - Respond to questions and engage in open-ended discussion (3%)

Getting Started

To help you get started with your project:

- Review the course materials and identify areas that interest you most
- Consider your team's strengths and interests
- Start with a well-defined, achievable scope
- Plan for incremental development and testing
- Utilize available computing resources (Google Cloud, Colab)
- Schedule office hours with instructors and TAs for guidance

Remember that the best projects often start simple and evolve based on what you learn during implementation. Focus on building a solid foundation and then iteratively improve your system.

Related Materials and Resources

Below are some useful resources to help you with your course project:

- **Basic Knowledge**

  - How to do research
  - Lessons for research
  - Other Knowledge
- **Open-source Robotics Hardware**
  - LeRobot
- **Robotics Simulators**
  - Isaac Sim - NVIDIA's scalable robotics simulation platform
  - SAPIEN - PartNet-Mobility dataset and simulation environment
  - MuJoCo - Fast and accurate physics simulator
  - Genesis - Open-source robotics simulation platform
- **Example Projects**
  - Particle-Grid Neural Dynamics for Learning Deformable Object Models from RGB-D Videos - Learning-based world model for deformable object manipulation
  - PhysTwin: Physics-Informed Reconstruction and Simulation of Deformable Objects from Videos - Physics-based world model for deformable object manipulation
  - D3Fields: Dynamic 3D Descriptor Fields for Zero-Shot Generalizable Rearrangement - Feature field for robotic manipulation
  - VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models - Foundation model for robotic manipulation
  - 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing - Tactile sensing for robotic manipulation